Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding

Lewei Xu[1] **Yihao Ding**[1*] **Zihan Xu**[2] **Siwen Luo**[1] **Daochang Liu**[1] **Wei Liu**[1]

[1]The University of Western Australia, Perth, Australia

[2]The University of Melbourne, Melbourne, Australia

[*]*Correspondence:* yihao.ding@uwa.edu.au

Abstract

Documents spanning tens to hundreds of pages are pervasive in real-world workflows, from scientific papers to financial reports and technical manuals. In this setting, answer-relevant evidence is sparse and distributed across pages, and the document itself often exceeds a backbone MLLM's context window, making MP-VRDU a distinct problem from single-page VRDU. We survey MP-VRDU through the lens of evidence management, the organising principle that distinguishes multi-page from the single-page setting: the problem of deciding how document-scale evidence is selected and reasoned over when it cannot all be held at once. We introduce a taxonomy spanning four architectural families, multimodal representation, inference-time adaptation, and training strategies for cross-page evidence. We further consolidate MP-VRDU datasets and outline open challenges in efficiency, faithfulness, evidence pipelines, supervision integrity, and reproducible evaluation that define the frontier.

1 Introduction

Visually-rich documents (VRDs) such as scientific papers, financial reports and technical manuals are pervasive in real-world workflows, and understanding them is a long-standing problem at the intersection of NLP and computer vision. Much of this work centres on the single-page (SP) setting. Real documents, however, often span dozens of pages, and answering a question about one may require combining a figure, a table, and a passage located many pages apart. Multi-page visually rich document understanding (MP-VRDU) is therefore not a simple extension of SP-VRDU, since total document length frequently exceeds the backbone MLLM's context window, evidence is sparsely distributed across pages, and document structure such as section hierarchies and cross-references cannot be recovered from per-token layout signals alone.

MP-VRDU has seen rapid growth in recent years, with new architectures, training strategies, and inference methods emerging. Across these lines of work, the reasoning engine is almost always a general-domain MLLM; what varies is the evidence-management machinery built around it to handle documents whose length and structure exceed what the MLLM can process directly. Existing work has explored joint multi-page encoding (Tito et al., 2023), adaptation of pretrained MLLMs (Duan et al., 2025b), retrieval-augmented pipelines that decouple evidence selection from generation (Cho et al., 2024), and adaptive controllers that gather evidence iteratively (Sun et al., 2025). However, this body of work has developed along largely parallel lines, with limited cross-family comparison and inconsistent terminology, and few works explicitly distinguish what is genuinely MP-distinctive from what is inherited from the single-page setting. As a result, the field lacks a unified view of the design choices that matter most in multi-page settings, motivating the need for systematic organisation.

Prior surveys have charted VRDU from complementary angles: OCR-centric deep-learning pipelines of the pre-MLLM era (Subramani et al., 2020), information extraction over visually rich documents (Ding et al., 2026a; Gbada et al., 2025), and more recent treatments covering the shift to MLLM-based architectures (Ding et al., 2026b) and multimodal retrieval-augmented generation for scaling beyond a single context window (Gao et al., 2026). Several of these include multi-page systems among the work they cover, but none treats multi-page understanding as a distinct problem with its own design axes.[1]

This survey provides a systematic account of the emerging MP-VRDU frontier. We define MP-VRDU as a problem distinct from SP-VRDU and

[1]Comparison table is available in Appendix A.

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing, pages 20159–20187
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Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VT5 (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2026)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	-	CoT, HD	-
Leopard (2025)	TMLR	No	SigLIP-SO400M	Mult. (La/Mi)	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B	V	-	HD	-
Docopilot (2025b)	CVPR	No*	InternViT-300M	Mult. (In)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2-7B	V	-	-	-
DocSLM (2026)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025)	CVPR	Yes	-	Mult. (La/Qw)	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2025)	MMAsia	No	Pix2Struct	Qwen2.5-VL-3B	V	D	-	-
DFVC (2026)	MDPI	No	VisRAG, ColQwen	Qwen2-VL-2B	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Mult. (Qw/In/Id)	V	D	-	-
MM-R5 (2025)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025b)	EMNLP	No	ColPali	Mult. (Qw/DS/Lv)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	CVPR	No	ColPali, Qwen3-VL ViT	Mult. (Qw/GL/Ge)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Mult. (In/Pa/Ph)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	ACM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024c)	AAAI	Yes	-	Mult.	T, L	S, IR, Nav	-	1A
MHier-RAG (2026)	preprint	Yes	Qwen-VL-Plus	Mult. (Qw/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Mult. (Qw)	T, V, L	D	CoT, HD	-
MultiDocFusion (2026)	ACL	Yes	-	Mult. (Qw/La/Mi)	T, V, L	D	HD	-
M2RAG (2025a)	NeurIPS	Yes*	VisRAG-Ret	Mult. (Qw)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Mult. (La/Qw)	T, V	J	-	5A
PDF-WuKong (2026)	IJCV	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Mult. (Qw/Ge/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	ACL	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025a)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Mult. (Qw/MM/In)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2025b)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B	V	D, IR, QR	ReAct, HD	1A
DocLens (2026)	ACL	Yes	–	Mult. (GPT/Ge/Cl)	T, V, L	J	CoT	4A, SA
DMAP (2026)	WWW	Yes*	ColPali	Mult. (GPT/Qw)	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	–	Mult. (GPT/Ge/Cl)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	ACL	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Mult. (Qw)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025a)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Mult. (Qw/La/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

Table 1: Survey of MP-VRDU models grouped by architecture. T = text; V = visual; L = layout. CoT = Chain of Thought; ReAct = Reason + Act; PE = Plan-Execute; QD = Question Decomposition; HD = Hierarchical / Coarse-to-Fine. S = Sparse; D = Dense; J = Joint; IR = Iterative Retrieval; QR = Query Reformulation; Nav = Navigation. nA = n agents; SR = Self-Refine; SV = Self-Verify; SC = Self-Consistency; SA = Sampling-Adjudication. OCR column: Yes* = not OCR-dependent; No* = OCR-derived supervision explicitly used during training. *Mult.* denotes a system evaluated on multiple backbones: Qw = Qwen; Ge = Gemini; GL = GLM; Cl = Claude; DS = DeepSeek; In = InternVL; Id = Idefics; La = LLaMA; Lv = LLaVA; Mi = Mistral; MM = MiMo; Ph = Phi; Pa = PaliGemma. '–' indicates not reported.

organise systems through an evidence-management perspective across four architectural families (§3), multimodal representation under document-length constraints (§4), and inference- and training-time strategies for cross-page evidence (§5–§6). We further consolidate datasets and evaluation practices and identify open challenges that define the MP-VRDU frontier (§7–§8).

2 Methodology and Scope

We cover MP-VRDU systems published from 2023 across major machine learning venues, together with arXiv preprints introducing multi-page-specific techniques. Candidates were identified through keyword search through forward- and backward-citation sweeps from an anchor set of widely-cited models and benchmarks, then manually screened for substantive multi-page contributions such as novel architectures, retrieval mechanisms, or training strategies. The dataset catalogue covers datasets used by the surveyed systems, plus a small number of relevant but underused ones. Rather than a systematic scoping review aiming at exhaustive retrieval, we prioritise representative coverage of the design space, since MP-VRDU is evolving rapidly and its terminology remains in flux. Full surveyed corpus details, search terms and inclusion criteria are available in Appendix B.

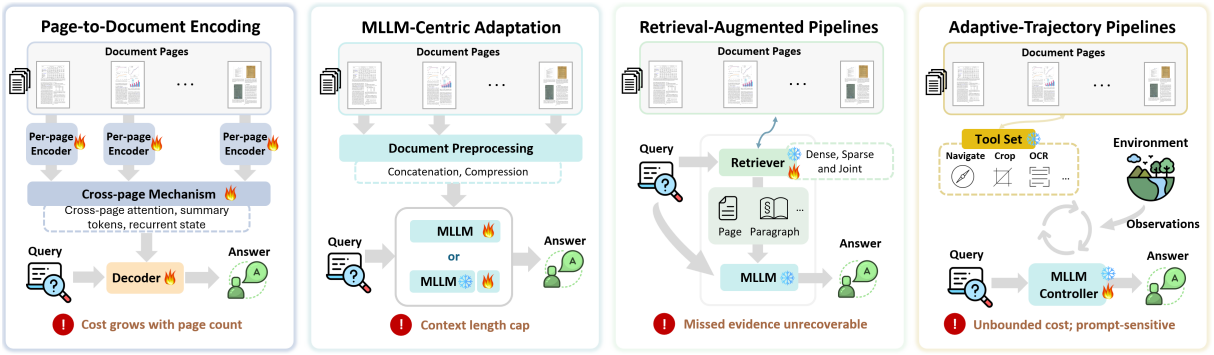


Figure 1: Illustration of MP-VRDU Architectures. Snowflake indicates component typically frozen; flame indicates component typically trained; both indicates even split across surveyed architectures.

3 MP-VRDU Architecture Overview

The defining problem in MP-VRDU is evidence management, and how a system manages cross-page evidence is fixed by its architectural design, which determines **how evidence is selected, preserved, and exposed for reasoning**. We therefore organise the field by architecture rather than by the OCR dependency used in prior surveys (Ding et al., 2026b), into four families. *Page-to-document encoding models* jointly encode pages with cross-page representations (§3.1). *MLLM-centric adaptation models* adapt MLLMs to multi-page inputs within context and token budgets (§3.2). *Retrieval-augmented pipelines* retrieve relevant evidence before generation (§3.3). *Adaptive-trajectory pipelines* iteratively inspect evidence using an LLM controller (§3.4). Reported performance across the four families on principal MP-VRDU benchmarks is provided in Appendix E.1 (Table 5).

3.1 Page-to-Document Encoding Models

Page-to-document encoding models process each page through a shared layout-aware encoder such as LayoutLMv3 (Huang et al., 2022) that fuses OCR text, layout, and visual evidence within the page, then connect per-page representations through a cross-page exchange mechanism in a single forward pass, such as per-page summary tokens (Tito et al., 2023), document-global attention (Blau et al., 2024), recurrent memory (Dong et al., 2024), or sparse cross-page attention (Borchmann et al., 2025). By preserving page identity and reading order within the architecture, this design provides a strong inductive bias for documents with regular templates such as forms and reports. However, cost grows with the number of pages held in the joint pass, capping total document length at the encoder's capacity, and page-to-document compression discards fine-grained evidence as input grows. This restricts the family to documents of shorter length.

3.2 MLLM-Centric Adaptation Models

MLLM-centric adaptation models process the document inside a pretrained MLLM whose context window absorbs cross-page information directly, with adaptation ranging from training lightweight modules such as projectors (Tanaka et al., 2024) and adapters (Zhu et al., 2025) to fine-tuning the backbone itself (Duan et al., 2025b; Xiong et al., 2026), often with visual-token compression to keep long documents within budget (Hu et al., 2025). The family inherits the backbone's instruction-following and reasoning and avoids the overhead of external retrieval. Its cost is bounded by the context window: a document longer than the window cannot be processed at all, and performance already degrades well before that point (Jia et al., 2025). Reliable text recovery depends on high-resolution input that token compression compromises, and document structure must be recovered implicitly rather than supplied through dedicated tokens (Hannan et al., 2026). Combined with attention dilution over long inputs (Liu et al., 2024), these limits motivate families that move evidence selection out of the backbone.

3.3 Retrieval-Augmented Pipelines

Retrieval-augmented pipelines decouple evidence selection from generation along a control flow fixed at design time: a retriever selects relevant evidence units, such as chunks, pages, or graph nodes, and the generator answers conditioned on this compact subset (Cho et al., 2024; Tanaka et al., 2025). This separation scales to documents

that exceed what page-to-document and MLLM-centric models can hold at once, since generator cost is capped at the retrieved units rather than tracking document length. The family spans single-pass retriever-generator systems (Cho et al., 2024) to multi-component pipelines composing several LLM-instantiated "agents" in specialised roles (Han et al., 2025; Duan et al., 2025a), but they all execute a predetermined sequence of steps. Its binding constraint is that evidence missed or poorly ranked at retrieval is unrecoverable at generation, so questions requiring aggregation of several weakly relevant pages frequently fail (Wu et al., 2025b), and upstream errors in chunking, OCR, or modality alignment propagate directly into the answer. The retrieval stage is itself an added cost, scaling with the document rather than with the retrieved subset, though it is cheap compared to answer inference.

3.4 Adaptive-Trajectory Pipelines

Adaptive-trajectory pipelines instantiate the agent paradigm of Plaat et al. and Wang et al., in which an LLM controller reasons, acts on an external environment, and conditions each action on observed outcomes rather than executing a fixed plan (Zheng et al., 2026; Sun et al., 2025). Unlike the fixed-role agents of §3.3, the controller's trajectory is not predetermined: it selects each evidence-gathering action at inference time, surfacing as calls to external retrievers, parsers, or OCR engines (Wang et al., 2025a; Yu et al., 2025), or as navigation over a prebuilt document representation (Jain et al., 2025; Fu et al., 2026). Because the trajectory is constructed rather than fixed, compute scales with question difficulty rather than document length, which lifts the recall ceiling of single-pass retrieval. The action sequence also becomes an inspectable audit trail. The cost of this flexibility is controllability: inference cost is hard to bound, and controllers may commit early to wrong evidence and continue down the wrong trajectory without surfacing the error, but the trade-off is worthwhile, since unbounded cost is recoverable through engineering whereas missed evidence is not.

4 Multi-Page Multimodal Representation

Every evidence-management decision is enacted over a specific modality, and its cost depends on how that modality is represented under the tension between page coverage and reading fidelity. This section reviews text (§4.1), visual (§4.2), and cross-page structural (§4.3) modelling under this pressure; fusion is deferred to Appendix C.1.

4.1 Text Modality

Text in MP-VRDU presents four bottlenecks. **Volume** across pages frequently exceeds the MLLM context, addressed by aggressive compression into a fixed budget (Borchmann et al., 2025; Dong et al., 2024), by native long-context training that widens the input window (Duan et al., 2025b), or by indexing chunks and passing only selected text to the reader (Xie et al., 2026; Zhang et al., 2024); each shifts the lossy choice to a different stage. **Continuity** breaks when paragraphs, tables, and sections span page boundaries; it is preserved through document-level layers over per-page representations (Tito et al., 2023; Blau et al., 2024) or structure-aware chunking and adjacent-page expansion at retrieval time (Shin et al., 2026; Qian et al., 2026), each restoring local context at the cost of token-level detail or query relevance. **Noise** compounds as OCR errors accumulate across the document, dampened upstream through layout-aware pipelines or downstream by treating low-confidence tokens as soft cues (Hannan et al., 2026), yet propagation through retrieval indexes remains uncharacterised. **Heterogeneity** in text density receives the least attention, with adaptive page selection (Li et al., 2025) the rare question-conditioned exception in a setting otherwise dominated by fixed-budget allocation.

4.2 Visual Modality

Visual evidence preserves signals text extraction loses, but its computational cost introduces three bottlenecks. **Resolution** cannot be maintained at full document length, since token cost scales super-linearly under dynamic cropping (Zheng et al., 2026); it is managed by token-reduction modules that keep every page available at lower fidelity (Hu et al., 2025; Yu et al., 2024; Jia et al., 2025), thumbnail-based browsing with selective high-resolution inspection (Zheng et al., 2026; Jain et al., 2025), or high-fidelity reading of only retrieved pages that inherits the selector's recall ceiling (Zhu et al., 2026; Yan et al., 2026). **Multi-signal competition** forces text-bearing pixels, non-textual evidence, and layout cues to share one per-page budget; visual RAG retains patch-level granularity but ranks by layout similarity rather than answer-region confirmation (Cho et al., 2024;

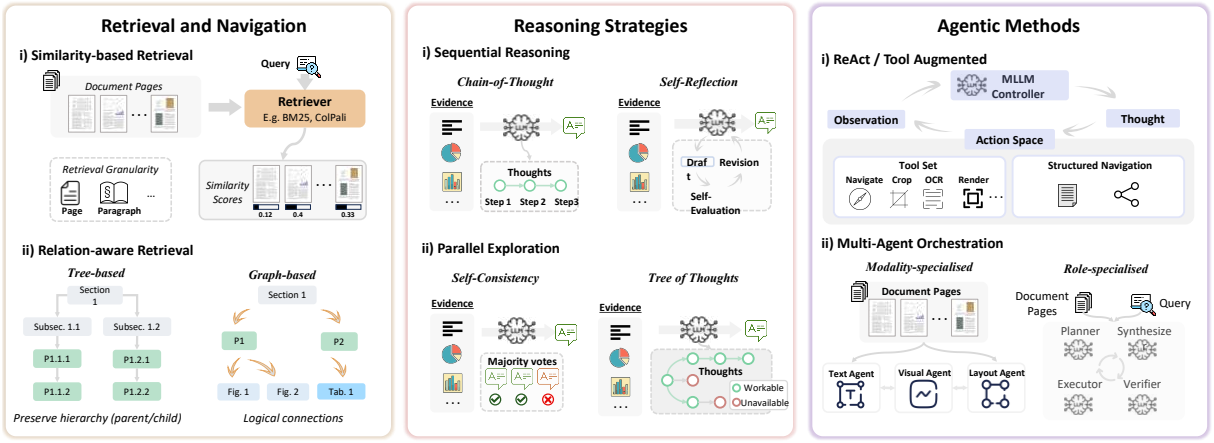


Figure 2: Illustration of Inference-Time Adapatation Strategies

Tanaka et al., 2025), element-level cropping preserves fine structure at the cost of detector dependence (Zhu et al., 2026), and OCR-augmented visual reading delegates text recovery so the visual budget can prioritise non-textual evidence (Hannan et al., 2026). **Heterogeneity** from varying visual complexity remains underexplored; evidence-guided allocation (Yan et al., 2026) is one of the few question-conditioned approaches.

4.3 Structure Modality

Structure turns isolated page observations into document evidence. Per-token layout remains relevant at MP scale (Zhu et al., 2025), but the MP-distinctive difficulty is structural relations whose scope crosses pages. **Hierarchy** is the foundational problem, since sections, headings, and reading order organise the document as a whole rather than any individual page. Explicit parsers recover this as outlines or hierarchical indices (Shin et al., 2026; Sun et al., 2025; Fu et al., 2026), OCR-free backbones learn it end-to-end from page images (Hu et al., 2025), and per-segment layout tokens encode each region's spatial arrangement without adding tokens to the context (Zhu et al., 2025). **Cross-page references** demand resolving pointers such as "see Figure 3" to their referents on other pages, rather than ranking by surface similarity; graph- and map-based methods represent these references as edges between document elements (Wu et al., 2025b; Fu et al., 2026), at the cost of edge quality bounded by extraction heuristics. **Spanning elements** such as multi-page tables require representations that bridge page boundaries without conflating distant content; hierarchical indexing and context-fused page embeddings preserve parent context (Gong et al., 2026; Qian et al., 2026).

5 Inference-Time Adaptation Strategies

Inference-time adaptation improves MP-VRDU performance by reshaping the inference pipeline around a frozen MLLM backbone rather than modifying its parameters, suiting a setting where supervised data is scarce and backbones must remain exchangeable. We therefore organise them by the evidence-management decision they implement: selecting evidence (§5.1), reasoning over assembled evidence (§5.2), and iteratively acquiring further evidence (§5.3).

5.1 Retrieval and Navigation

Retrieval narrows long documents to compact evidence units for downstream reasoning. Existing methods either rank evidences independently by query *similarity* or traverse *relation-aware* structures such as section hierarchies and cross-reference graphs.

Similarity-based Retrieval. Similarity-based retrieval scores evidences independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM-25 serve as baselines or auxiliary channels, while dense retrieval in MP-VRDU typically uses vision-language late-interaction encoders such as ColPali and ColQwen that align token-level query vectors with patch-level page embeddings, preserving fine-grained alignment between query terms and page regions (Cho et al., 2024; Jain et al., 2025). Granularity varies from chunks to whole pages, with adaptive selectors adjusting the per-query retrieval count when evidence density is uneven (Li et al., 2025). A second reranking stage commonly refines

an initial dense recall through LLM-based scoring over candidate evidences or page summaries, optionally under persistent working memory that carries cross-query context across iterations (Zhang et al., 2024; Xu et al., 2025). Independent scoring handles paraphrase and cross-modal queries well, but cannot follow cross-page references or multi-hop dependencies that no single passage carries.

Relation-aware Retrieval. Relation-aware methods retrieve along explicit document structure rather than scoring evidences independently against the query. Tree-based variants extract section hierarchies and outlines, grouping evidences along semantic boundaries so retrieved chunks preserve parent context, with some routing queries through dual in-page and cross-page indices to recover both local and global evidence (Shin et al., 2026; Sun et al., 2025; Gong et al., 2026). Graph-based variants treat headings, entities, and cross-references as graph nodes with typed edges, allowing controllers to traverse from one passage to related ones rather than relying on similarity alone; extensions add logical relevance scoring or query-conditioned graph construction to bias traversal toward inferentially connected nodes (Wang et al., 2024c; Wu et al., 2025b; Zhang and Wu, 2026). Structure-based methods recover evidence links that similarity scoring misses, at the cost of dependence on parser quality and traversal cost that grows with the multi-hop count required to answer the question.

5.2 Reasoning Strategies

Reasoning strategies act over a single evidence buffer assembled prior to the reasoning call, which may be the full document representation, a retrieved subset, or evidence from an agentic step. They range from purely prompted interventions to strategies requiring changes to decoding, sampling, or control flow, but none updates model parameters. We cover the four mainstream reasoning paradigms; the coarse-to-fine pattern that recurs across many surveyed systems is treated separately in Appendix C.2.

Sequential Reasoning. Sequential strategies extend a single reasoning thread over the evidence buffer, either elaborating it step by step or revising it across rounds. *Chain-of-Thought (CoT)* (Wei et al., 2022) elicits intermediate reasoning steps before the final answer. In MP-VRDU it serves two purposes: binding evidence drawn from different pages into a single chain, and decomposing multi-hop questions into single-hop steps that each rest on one page (Jia et al., 2025; Xu et al., 2025). Its contribution is therefore confined to the fidelity stage of the pipeline, since a chain can only reason over the evidence already assembled and cannot recover what retrieval missed. *Self-Reflection* (Reflexion-style) (Shinn et al., 2023) extends the single thread with iterative revision, having the model evaluate its own draft answer and revise across rounds, optionally against a verifier or critic (Jain et al., 2025; Wang et al., 2025a). This reduces single-pass error, but each round adds inference cost, and the reliability of the self-evaluation signal bounds how much revision can recover.

Parallel Exploration. Parallel strategies produce several reasoning paths over the same evidence buffer and combine them into a single answer, either by aggregating independent samples or by searching over branches. *Self-Consistency* (Wang et al., 2023) samples multiple chains from the same prompt and selects the majority answer. Variants change how the selection is made: sampling-adjudication replaces the vote with an LLM judge that scores candidates by reasoning quality (Zhu et al., 2026), while parallel-pipeline designs enforce consistency across modality-specific chains rather than across samples of one chain (Suri et al., 2025). All three trade inference cost for robustness against single-chain failure, and cost scales with the number of samples drawn. *Tree-of-Thoughts (ToT)* (Yao et al., 2023a) instead explores reasoning branches with backtracking, which requires a state-evaluation signal to judge partial paths. Such a signal is difficult to define over document evidence, and ToT has seen little uptake in MP-VRDU, where the dominant difficulty is evidence sparsity rather than search-space branching.

5.3 Agentic Methods

Agentic methods distribute the MP-VRDU workflow across one or more LLM-instantiated controllers, either iterating evidence acquisition under inference-time control or coordinating across specialised roles. We note that *agentic* is used ambiguously in the MP-VRDU literature, covering both systems whose controller selects actions at inference time, following the framing by Plaat et al. and Wang et al., and systems composed of multiple LLM-instantiated agents in fixed roles; we treat both senses here, following the architectural distinction in §3.3–§3.4.

ReAct and Tool-Augmented Reasoning. ReAct prompting (Yao et al., 2023b) augments the action space with language thoughts, interleaving thought-action emissions in a single stream whose observations feed the next step, letting a single controller acquire evidence during inference rather than reasoning over a fixed buffer. The action space takes two forms in MP-VRDU. Document-side tool calls invoke external retrievers, parsers, or OCR engines, with each round concentrating on a chosen page or region (Sun et al., 2025; Wu et al., 2025a), lifting the single-step recall ceiling of retrieval-augmented pipelines. Structured navigation instead operates over a pre-built document representation (see §4.3), with the controller selecting which node, section, or page to inspect next (Zheng et al., 2026). Adaptive-trajectory systems that lack the thought-action schema, such as those built on Reflexion-style iterative refinement (Jain et al., 2025; Fu et al., 2026), share the iterative control regime without being ReAct in the strict sense.

Multi-Agent Orchestration. Multi-agent orchestration distributes the workflow across multiple controllers whose interaction pattern, rather than a single monolithic controller, drives evidence gathering and answer synthesis. Each controller operates on a narrower input and clearer objective than a single backbone tasked with the entire workflow, mitigating the cognitive overload that arises when one model must locate, read, verify, and answer over the entire document (Yu et al., 2025). Two patterns recur. *Modality-specialised* decomposition assigns separate controllers to text and visual evidence and adds a synthesis controller that reconciles their outputs, preserving single-modality reading fidelity at the cost of a reconciliation step that lacks access to the original document (Han et al., 2025; Duan et al., 2025a). *Role-specialised* decomposition assigns controllers to procedural stages such as planning, execution, verification, and synthesis, with routing between stages when a downstream controller detects an error (Yu et al., 2025; Sun et al., 2025). Separating judgement from generation reduces the cognitive blind spots of self-correction, but introduces inter-controller compatibility costs and a recall ceiling bounded by upstream stages.

6 Training Strategies

VRDU training follows a stepwise pipeline established in the single-page setting (Ding et al., 2026b): self-supervised pretraining instils document reading through objectives such as masked information modelling and cross-modality alignment, instruction tuning aligns outputs with user queries, and supervised fine-tuning adapts the model to target tasks. MP-VRDU inherits this pipeline but concentrates its effort at the later stages, since document-scale pretraining corpora remain scarce and most systems adopt backbones already pretrained on single-page data, with few exceptions targeting cross-page objectives directly (Hu et al., 2025; Duan et al., 2025b). Fine-tuning instead carries the weight, scaled by LVLM-synthesised supervision (§7), with reinforcement learning increasingly layered above it. Training therefore internalises the evidence-management decisions of §5 into model parameters, supplying capabilities that prompting cannot: cross-page evidence selection (§6.1), document-scale reasoning over assembled evidence (§6.2), and reliable agentic acquisition (§6.3). Two cross-cutting techniques recur across all three (§6.4).

6.1 Trained Retrieval Components

Off-the-shelf encoders such as ColPali and ColQwen (Faysse et al., 2025) score each page independently, leaving retrieval dominated by local query-page similarity rather than document-level context. Retriever training has therefore explored three complementary directions. **Modality-alignment** methods align visual page embeddings with OCR-derived text representations, enabling page-image retrieval without late-interaction scoring at inference (Tanaka et al., 2025), trading index size for query-time efficiency. **Cross-page-fusion** methods enrich each page representation with neighbouring-page context (Qian et al., 2026), recovering distributed evidence without rebuilding the encoder, though the optimal neighbour window is corpus-dependent and rarely tuned per query. **Logical-relevance** methods train graph-based scorers that rank pages by their position in cross-page reasoning paths rather than embedding similarity (Wu et al., 2025b), addressing multi-hop questions whose evidence pages are individually low-scoring at the cost of inheriting the graph-extractor's failure modes. The three directions are not interchangeable: modality-alignment improves throughput, cross-page-fusion improves recall on adjacent evidence, and logical-relevance improves recall on inferentially-connected evidence.

6.2 Trained Reasoning Methods

Prompted reasoning leaves quality sensitive to prompt design and base-model behaviour, motivating training that internalises reasoning patterns into the backbone. **Trace-imitation** variants supervise the backbone on demonstration chains wrapped in fixed schema tags such as `<think>`, `<evidence>`, and `<answer>` (Yan et al., 2026), which provides structure that downstream verifiers can audit but inherits any biases or omissions in the teacher's chains. **Reward-driven** variants optimise the reasoning policy with Group Relative Policy Optimisation (GRPO) against rewards combining answer accuracy, evidence-page recall, and format compliance (Xiong et al., 2026; Xu et al., 2025), typically warm-started from a trace-imitation stage; the two are therefore sequential rather than alternative, with reward-driven training refining what imitation alone cannot. **Uncertainty-calibration** variants learn an aggregation head that selects across candidate outputs by calibrated uncertainty (Hannan et al., 2026), providing a learned self-consistency signal without test-time chain sampling, which suits parameter or compute budgets that preclude sampling-based approaches.

6.3 Trained Agentic Methods

Prompt-only agentic control is unreliable on smaller open-weight backbones, which cannot consistently follow ReAct schemas and select among tool calls. **Trajectory-distillation** variants clone agentic trajectories from a stronger teacher through SFT on think-action-observation turns (Zheng et al., 2026; Zhang et al., 2026), with closed-loop teacher collection essential because retrieval and fetching change subsequent observations; distillation is bounded by teacher quality and inherits the teacher's blind spots. **Reinforcement-learned** variants optimise the agentic policy directly against trajectory-level rewards combining answer correctness, evidence recall, and tool-use shape (Wang et al., 2025b; Lin et al., 2026), lifting the teacher-quality ceiling at the cost of training instability that multi-turn variants address through entropy-based stopping, similarity-weighted baselines, or per-role budget allocation. The surveyed evidence does not cleanly separate the two, since reported comparisons confound the training objective with backbone, reward design, and trajectory budget.

6.4 Cross-Cutting Training Techniques

Two techniques recur across retriever, reasoner, and agent training. **Parameter-efficient tuning** (PEFT) is dominant: full backbone fine-tuning is prohibitive at long-document context lengths and risks erasing pretrained capabilities, so the prevailing pattern freezes the backbone and trains lightweight modules through Low-Rank Adaptation (LoRA), projection heads, or task-specific adapters (Zhang et al., 2024; Qian et al., 2026; Zhang et al., 2026), with the choice of module driven more by compute budget than by target component. **Multi-stage curriculum training** addresses a different concern: several systems start on single-page objectives before introducing document-scale supervision (Hu et al., 2025; Hannan et al., 2026; Xiong et al., 2026), exploiting the fact that text reading and layout understanding transfer from single-page data while cross-page integration must be learned separately. Their recurrence across dissimilar systems indicates that cross-page integration does not emerge from single-page supervision alone.

7 Datasets

Multi-page documents impose a structural constraint absent from the single-page setting: per-page annotation does not scale, and multi-page corpora are scarce because long PDFs are harder to source, license, and parse than single pages. This constraint shapes all three dataset roles in MP-VRDU differently from SP-VRDU. **Pretraining** is the exception rather than the norm, confined to a few page-to-document and MLLM-centric systems that perform continued pretraining on existing backbones, since multi-page pretraining corpora remain small relative to the single-page archives that preceded them. **Fine-tuning** has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour rather than to scale the data itself. **Benchmarking** has stratified into three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks requiring cross-page integration, with retrieval and page-localisation increasingly scored alongside answer accuracy. The same scarcity of source documents that drives synthetic generation also concentrates fine-tuning and evaluation onto the same small pool of PDFs, a contamination risk we return to in §8. For full details see Appendix D.

8 Challenges and Future Work

A consistent pattern emerges across MP-VRDU systems: progress comes from moving evidence handling out of the backbone, through compression, retrieval, or iterative orchestration, but each shift relocates the bottleneck rather than removing it. Five challenges define the open frontier: three concern system capability (the evidence pipeline itself, faithfulness, and efficiency) and two concern data and evaluation (supervision integrity and evaluation scope).

Evidence Pipeline Bottleneck. MP-VRDU systems must reconcile high-resolution per-page reading with document-scale evidence aggregation, an objective that exceeds any MLLM's compute and context budget (Ma et al., 2024). Every architectural family takes a lossy compromise: compression silently drops fine-grained content and is typically fixed before the question is seen; retrieval bounds answer accuracy from above while broader retrieval introduces attention dilution; and iterative retrieval lifts this ceiling only partially, inheriting the recall of every tool it calls. Cross-page structure compounds this further, as none of the routes in §4.3 scales cleanly. Future work should explore query-conditional compression, retrieval that follows cross-references and logical relations, and learned cross-page structural representations.

Faithfulness and Abstention. Long documents amplify hallucination risk; when retrieval misses or compression discards the answer-bearing region, current systems still produce confident answers. Few surveyed systems support abstention, calibrated uncertainty, or citation-grounded generation, leaving downstream users without signals to distinguish supported answers from fabrications. Future work should integrate evidence-grounded decoding, learned abstention against unanswerable or unretrievable questions, and trajectory-level faithfulness audits that verify the cited evidence supports the final answer.

Efficiency and Deployability. Agentic and retrieval-augmented pipelines routinely issue tens of LLM calls per question, with token costs scaling jointly with document length and trajectory length. The surveyed literature reports almost no inference-cost analysis: end-to-end latency, token budgets per query, and accuracy–cost Pareto curves are largely absent, despite their centrality to enterprise deployment. Future work should treat compute as a first-class evaluation axis, develop efficient inference techniques specific to long documents such as cross-round caching, speculative decoding, and adaptive depth, and explore compact MP-VRDU models (Hannan et al., 2026) that operate under realistic on-device or privacy constraints.

Supervision Scarcity and Contamination. Per-page annotation does not scale to documents averaging tens of pages, so most systems rely on LVLM-synthesised QA over scraped PDFs (Tanaka et al., 2024; Jain et al., 2025; Xiong et al., 2026; Zhang et al., 2026). Compounding this, benchmark construction often reuses documents from earlier suites: MMLongBench-Doc draws documents from DUDE, SlideVQA, and ChartQA (Ma et al., 2024), while MP-DocVQA is built on DocVQA (Tito et al., 2023; Mathew et al., 2021). Together, MLLM-generated QA data (Deng et al., 2025) and benchmark-reused fine-tuning sets (Xiong et al., 2026; Duan et al., 2025b) risk contamination: models may train on benchmark-derived documents and be evaluated on overlapping data, sometimes judged by the same MLLM family. Future work should adopt held-out documents, declare cross-suite overlap, and separate generators from evaluators. More details and overlap matrix available in Appendix E.3.

Evaluation Methodology and Scope. Beyond contamination, two further evaluation gaps persist. Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning; newer benchmarks score retrieval and page-localisation alongside VQA-style answers (Hui et al., 2024; Zhu et al., 2026), but the practice is not yet standardised, and agentic pipelines produce stochastic trajectories that further complicate reproducibility. Task, language, and domain scope also remain narrow. Real-world MP-VRDU spans summarisation and structured extraction beyond VQA, multilingual layouts beyond English, and domain-specialised settings such as medical, legal, and financial documents, as well as streaming inputs that arrive page-by-page. These dimensions are inherited from SP-VRDU (Ding et al., 2026b) rather than introduced by the multi-page setting and remain underexplored in both regimes. Future work should therefore standardise trajectory-level evaluation with retrieval and grounding metrics, report stochasticity bounds, and broaden coverage along these inherited dimensions.

Limitations

This survey covers multi-page VRDU models published before May 2026 and is bounded by the rapid pace of the field, with new architectures, training-free methods, and benchmarks appearing during preparation that may not be reflected here. Selection prioritises representative coverage of architectural and inference-time strategies over exhaustive enumeration. Cross-model performance comparisons are inherently limited since training distributions, benchmark coverage, and evaluation metrics are bespoke per system, and quantitative claims should be interpreted as indicative rather than definitive.

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Figure 3: MP-VRDU Taxonomy

Survey	Venue	Scope	Multi-Page Treatment	Organising Framing
A Survey of Deep Learning Approaches for OCR and Document Understanding (Subramani et al., 2020)	arXiv 2020	Pre-MLLM OCR pipelines	Not covered	Pipeline stage (detection, recognition, layout)
Deep Learning Approaches for Information Extraction from Visually Rich Documents: Datasets, Challenges and Methods (Gbada et al., 2025)	IJDAR 2025	Information extraction over VRDs	Not covered	Extraction task and method
Deep Learning based Visually Rich Document Content Understanding: A Survey (Ding et al., 2026a)	AI Review 2026	Content understanding over VRDs	Not covered	Task formulation, architecture, pretraining
A Survey on MLLM-based Visually Rich Document Understanding: Methods, Challenges, and Emerging Trends (Ding et al., 2026b)	ACL 2026	MLLM-era VRDU	In passing	OCR dependency; architecture, training, benchmarks
Scaling Beyond Context: A Survey of Multimodal Retrieval-Augmented Generation for Document Understanding (Gao et al., 2026)	ACL 2026	Multimodal RAG for long documents	Partial (RAG techniques only)	Retrieval-augmented generation
This survey	EMNLP 2026	**Multi-page VRDU**	**Distinct problem**	**Evidence management**

Table 2: Comparison of this survey with prior VRDU surveys. Prior surveys either predate the multi-page literature or address it only within a broader single-page or extraction-focused scope. This survey is the first to treat multi-page VRDU as a distinct problem organised around evidence management.

A Comparison with Prior Surveys

Table 2 contrasts this survey with the five prior VRDU surveys closest to our scope. The comparison is along three axes. **Scope** records the era and problem each survey targets, from pre-MLLM OCR pipelines to the recent MLLM and retrieval-augmented literature. **Multi-page treatment** records whether multi-page understanding is treated as a distinct problem with its own design axes, addressed only in passing within a broader scope, or left out. **Organising framing** records the lens each survey uses to structure the field. Prior surveys either predate the multi-page literature or treat it as one topic within a single-page or extraction-focused scope. This survey is the first to define MP-VRDU as a distinct problem, to organise it around evidence management, and to cover the four architectural families, modality pressures, inference-time and training strategies, and datasets specific to the multi-page setting.

B More Methodology Details

B.1 Search keywords.

Queries combined a multi-page cluster with a task or method cluster. Multi-page terms included "multi-page document", "long document", "long-context document", "cross-page reasoning", "page-level retrieval", "document-scale evidence", and "hundred-page". Task and method terms included "document visual question answering", "visually rich document understanding", "document VQA", "document agent", "visual document retrieval", "document RAG", "hierarchical document transformer", and "long-context vision-language model". Benchmark-name probes ("MP-DocVQA", "MMLongBench-Doc", "LongDocURL", "DUDE", "SlideVQA", "M3DocVQA", "DocBench", "M-LongDoc") were used as anchors for forward-citation sweeps to capture systems that report against multi-page evaluation suites.

B.2 Inclusion and exclusion criteria.

A paper was retained if it (i) proposed a multi-page model, framework, system, or pipeline rather than a benchmark or dataset alone, and (ii) reported quantitative results on at least one multi-page benchmark, with cross-page evidence handling as an explicit design contribution. Excluded categories were single-page-only systems benchmarked solely on DocVQA, InfographicVQA, ChartQA, FUNSD, or CORD; survey, position, and opinion papers; generic OCR or table-extraction tools without VQA or reasoning evaluation; long-context LLM benchmarks where documents are

Model	Venue	Multi-Page Contribution
Page-to-Document Encoding Models		
Arctic-TILT (2025)	ACL	Layer-wise tensor-product text-image fusion with chunked sparse attention, serving up to 389k tokens on a single GPU
GRAM (2024)	CVPR	Interleaves page-level and document-level attention sub-layers with learnable global Doc tokens, avoiding quadratic concatenation cost
Hi-VT5 (2023)	PR	Learnable [PAGE] tokens summarise each page; decoder consumes the concatenated page tokens, with answer-page prediction for explainability
RM-T5 (2024)	DAS	Recurrent memory tokens propagate context page-by-page; per-page memories concatenated for decoding
MLLM-Centric Adaptation Models		
CoR (2026)	preprint	Four-stage Chain-of-Reading paradigm with a caption-reconstruction objective for figure and table grounding
Docopilot (2025b)	CVPR	Native multi-page MLLM trained on Doc-750K cross-page proxy tasks with context-extension engineering
DocR1 (2026)	AAAI	EviGRPO adds an evidence-page F1 reward over structured think/evidence/answer outputs
DocSeeker (2026)	CVPR	Analysis-Localization-Reasoning paradigm with page-identifier grounding and evidence-guided resolution allocation
DocSLM (2026)	CVPR	Hierarchical compressor fixes 576 tokens per page; streaming segments with entropy-based abstention for edge deployment
InstructDoc (2024)	AAAI	Document-former fuses OCR, layout, and visual features into learnable tokens, with pages encoded in parallel
LayTokenLLM (2025)	CVPR	Per-segment layout token shares its segment's position ID, adding layout without consuming context
Leopard (2025)	TMLR	Adaptive high-resolution multi-image encoding under a global token budget with pixel-shuffle compression
mPLUG-DocOwl2 (2025)	ACL	DocCompressor cross-attends global low-resolution queries to sub-image features, compressing each page to 324 tokens
Texthawk2 (2024)	preprint	Two-step resampling and rearrangement for 16× visual-token compression with a coordinate-token detection head
URaG (2026)	AAAI	Cross-modal retrieval module at an early Transformer layer discards non-retained pages before deeper layers
Retrieval-Augmented Pipelines		
AVIR (2025)	MMAsia	Adaptive page selector uses score-distribution clustering to set per-query retrieval depth
CREAM (2024)	ACM	Coarse-to-fine retrieval with iterative reranking, paired with attention-pooled multi-page visual features
DFVC (2026)	MDPI	Adapter predicts gating weights over neighbouring-page embeddings to fuse cross-page context into page representations
DREAM (2025)	MM	Learning-to-rank fusion of embedding similarity and keyword-guided MLLM scores, with cross-page mixture-of-experts routing
KGP (2024c)	AAAI	Knowledge graph over passage and structural nodes traversed by an LLM agent to gather supporting facts
M2RAG (2025a)	NeurIPS	Dual-tower sparse and dense retrieval with text, visual, and fusion agents performing consistency checking
M3DocRAG (2024)	preprint	ColPali late-interaction page retrieval with approximate indexing, scaling to 40k+ pages
MDocAgent (2025)	preprint	Five-stage pipeline with parallel text and image retrieval and specialised modality agents
MHier-RAG (2026)	preprint	Combines flattened in-page indexing with a recursively summarised cross-page tree for multi-granularity retrieval
MLDocRAG (2026)	preprint	Chunk-Query Graph of LVLM-generated queries links chunks for retrieval by query-to-query similarity
MM-R5 (2025)	AAAI	Two-stage reranker training with structured reasoning chains and rank-discounted GRPO rewards
MoLoRAG (2025b)	EMNLP	Traverses a page graph combining semantic and VLM-judged logical relevance to reach weakly-similar evidence
MultiDocFusion (2026)	ACL	LLM-constructed document hierarchy tree drives depth-first hierarchical chunking
PDF-WuKong (2026)	IJCV	End-to-end sparse sampler jointly trained with the LLM selects top-K text and image evidence
RAG-DocVQA (2025)	ICDAR	Systematic comparison of textual and visual RAG variants for multi-page DocVQA
Self-Attention Scoring (2024)	ICDAR	Self-attention scoring module ranks independently encoded pages, extending evaluation to 793-page documents
SLEUTH (2025)	CVPR	Training-free clue discovery and page screening keep effective context fixed as retrieval depth grows
SV-RAG (2025)	ICLR	Dual LoRA adapters over one MLLM backbone serve both late-interaction page retrieval and answering
VDocRAG (2025)	CVPR	Compresses each page image to a single dense embedding via self-supervised compression pretraining tasks
VisDoMRAG (2025)	NAACL	Parallel textual and visual RAG chains reconciled by consistency-constrained modality fusion
Adaptive-Trajectory Pipelines		
DMAP (2026)	WWW	Hierarchical document map traversed by tri-path retrieval with a reflective loop for insufficient evidence
Doc-V* (2026)	ACL	Thumbnail overview then alternating retrieval and high-resolution page fetch, trained by SFT then GRPO
DocAgent (2025)	EMNLP	Hierarchical XML outline navigated by an actor agent with five tools, plus a reviewer and persistent memory bank
DocDancer (2026)	preprint	End-to-end trained agent with only search and read tools, learned from synthesised exploration trajectories
DocLens (2026)	ACL	Page navigator and element localiser zoom to fine-grained evidence; sampling-adjudication selects among candidates
DocReact (2025a)	ACL	Frames iterative retrieval as information-gain maximisation, halting when residual entropy stops decreasing
MACT (2025)	preprint	Four specialised agents with independent judgment routing and agent-wise adaptive test-time scaling
MM-Doc-R1 (2026)	ACL	Planner-seeker-answer workflow trained by similarity-weighted policy optimisation for multi-turn RL
SimpleDoc (2025)	EMNLP	Dual-cue retrieval over embeddings and page summaries, with a memory-carrying reasoner that reissues refined queries
ViDoRAG (2025a)	EMNLP	Gaussian-mixture adaptive top-K retrieval with seeker, inspector, and answer agents over multi-scale views
VRAG-RL (2025b)	NeurIPS	Visual perception action space lets the agent crop and zoom regions, trained with retrieval-efficiency rewards

Table 3: The 46 surveyed MP-VRDU systems, grouped by architectural family, with the multi-page contribution motivating each system's inclusion. Venue indicates the publication venue at time of writing.

one input type among many; and dataset-only contributions, which are catalogued separately in Appendix D. Peer-reviewed venue publications and arXiv preprints were assessed against the same substantive criteria, since recent MP-VRDU progress has appeared predominantly in preprint form; venue status is recorded per entry in Table 1. Table 3 lists the surveyed systems grouped by architectural family, with the multi-page contribution on which each was included.

B.3 Knowledge extraction and verification.

Each retained paper was converted to markdown and processed through LLM-assisted key-information extraction with evidence-span location, populating per-model fields covering architectural family, OCR dependency, vision encoder, LLM backbone, adaptors and projectors, page-rendering resolution, training stages, datasets used per stage, and reported scores on each benchmark. Every extracted field was manually verified against the source paper before being added.

B.4 Dataset catalogue.

The dataset catalogue in Appendix D was built primarily from the pretraining, fine-tuning, and benchmarking datasets named by every paper in the surveyed corpus, supplemented with a small set of datasets the corpus does not yet adopt but that cover MP-VRDU-relevant settings such as long scientific documents, multi-page tables, and cross-page reasoning suites, included to surface datasets whose adoption may strengthen future cross-system comparison.

C More Framework Details

C.1 Multimodal Fusion

In multi-page VRDU, the central challenge is not only multimodal fusion itself, but determining when and how text, visual, and layout information should be selectively fused across large collections of pages under strict computational constraints.

Early Fusion. Text, visual, and layout features are combined before or during document encoding. In page-to-document transformers, fusion is local: OCR tokens, bounding boxes, visual patches, and question tokens are combined within each page and then aggregated through page summaries (Tito et al., 2023), document-level tokens (Blau et al., 2024), recurrent memory (Dong et al., 2024), or sparse text-image-layout attention (Borchmann et al., 2025). Backbone-centric MLLMs apply early fusion at projection time, mapping visual or multimodal features into the LLM token space through adapters, projectors, or document-former modules (Tanaka et al., 2024; Duan et al., 2025b; Jia et al., 2025). The pattern preserves fine-grained multimodal alignment at the cost of joint computation over many pages.

Late Fusion. Modalities are kept separate during retrieval or inspection and combined only after candidate evidence has been selected. Retriever-generator systems identify relevant evidences via text chunks before visual generation (Zhang et al., 2024), perform page-image retrieval that keeps multimodal evidence fused at the page level (Cho et al., 2024), or combine parsed text, visual descriptions, parent pages, and cross-page summaries only for retrieved evidence (Gong et al., 2026). Agentic pipelines extend the pattern by inspecting text, image, table, and page-level evidence separately before final synthesis or adjudication (Han et al., 2025; Duan et al., 2025a; Zhu et al., 2026). The pattern scales to long documents but depends on retrieval and routing quality.

C.2 Coarse-to-Fine Reasoning

Coarse-to-fine reasoning is the most widely adopted structured-reasoning pattern in the surveyed corpus, appearing across MLLM-centric, retrieval-augmented, and adaptive-trajectory systems (Table 1). The pattern decomposes document-scale reasoning into successive stages of decreasing scope: an initial pass over the whole document or a broad candidate set identifies regions likely to contain answer-relevant evidence, and subsequent passes restrict attention to those regions at higher fidelity. Three implementations recur. **Page to region** variants first select candidate pages from the document, then reason over each candidate at higher fidelity or with more detailed analysis before answering (Shi et al., 2026; Xiong et al., 2026; Yan et al., 2026; Gong et al., 2026), with some realisations operating across discrete passes and others embedding the pattern within the layers of a single MLLM. **Retrieve, rerank, then read** variants extend the pattern across the retrieval stack, with an initial dense or sparse retrieval returning a broad candidate set, a reranker narrowing it through LLM-based scoring, and the generator reading only the final subset (Zhang et al., 2024; Xu et al., 2025; Shin et al., 2026; Zhang and Wu, 2026). **Plan then exe-**

cute variants instantiate the pattern at the controller level, first producing an evidence-acquisition plan that identifies which pages or regions to inspect, then executing the plan in sequence with iterative refinement (Zheng et al., 2026; Sun et al., 2025; Fu et al., 2026). Across all three, early stages operate over compact representations and late stages at full fidelity over a narrowed scope, making coarse-to-fine a natural fit for the cost structure of MP-VRDU, though errors at an early stage are unrecoverable at later stages since later stages see only what was admitted upstream.

D More Dataset Details

Datasets in MP-VRDU span three roles: pretraining (PT), supervised fine-tuning (SFT), and benchmarking (BM), with several datasets occupying more than one role across different systems. Per-dataset statistics are available in Table 4 and per-paper adoption in Table 10.

D.1 Pretraining

Pretraining is the exception rather than the norm in MP-VRDU. Only a handful of page-to-document and MLLM-centric systems pretrain or continue-train a backbone at all, while retrieval-augmented and agentic pipelines inherit pretrained vision-language backbones and skip the stage entirely. Two factors explain its scarcity. Document-scale pretraining is computationally expensive, and genuinely multi-page pretraining corpora are few, so most systems that do pretrain rely on single-page archives or repurpose single-page data. Early work used single-page OCR archives such as IIT-CDIP (Lewis et al., 2006) and OCR-IDL (Biten et al., 2022), which supply text-recognition supervision but no cross-page structure. The small number of genuinely multi-page corpora appeared only recently, as DocStruct4M (Hu et al., 2024) mixes single- and multi-page samples with structure-aware parsing supervision, and the purely multi-page MP-DocStruct1M (Hu et al., 2025), Doc-750K (Duan et al., 2025b), and bilingual Paper-PDF corpora (Xie et al., 2026) support multi-image continued pretraining over genuine document inputs. Where multi-page pretraining does occur, the objective has moved from text recognition toward structure-aware supervision, though the corpus as a whole shows that most cross-page capability is instilled at fine-tuning rather than pretraining.

D.2 Supervised Fine-tuning

Per-page human annotation does not scale to the page counts typical of MP-VRDU, so supervised fine-tuning has converged on two strategies that are often combined within a single system. The first strategy aggregates publicly released question-answer collections into broad instruction mixes that exercise heterogeneous question and answer types (Tanaka et al., 2024; Jia et al., 2025; Hu et al., 2025; Duan et al., 2025b; Ding et al., 2025). The second strategy curates a bespoke corpus for the system under study, typically synthesised by a strong vision-language model writing question-answer pairs, reasoning traces, or agent trajectories over scraped PDFs (Xie et al., 2026; Tanaka et al., 2025; Wu et al., 2025b; Zhang et al., 2026; Guo et al., 2026). Synthetic generation has become the dominant scaling path for fine-tuning data, and the underlying PDF pools are typically drawn from the same scientific archives, slide repositories, and benchmark document sources that also supply evaluation. Layered above supervised fine-tuning, a growing body of work applies reinforcement or preference learning with shaped rewards that credit correct evidence-page citation, faithful chain-of-thought structure, or appropriate tool use rather than final-answer accuracy alone (Xiong et al., 2026; Guo et al., 2026; Yan et al., 2026; Lin et al., 2026; Xu et al., 2025; Wang et al., 2025b). The supervision pools at this stage remain small relative to the supervised mixes that precede them, indicating that reinforcement signals are used for behaviour shaping; instilling evidence grounding, output formatting, and multi-step reasoning patterns, rather than for further scaling.

D.3 Benchmarking

Benchmarks have evolved through several tiers that have accumulated rather than replaced one another. Single-page extractive evaluation on DocVQA (Mathew et al., 2021), InfographicVQA (Mathew et al., 2022), and ChartQA (Masry et al., 2022) persists as a per-page baseline, with mid-length multi-page corpora such as MP-DocVQA (Tito et al., 2023), DUDE (Landeghem et al., 2023), and SlideVQA (Tanaka et al., 2023) extending the same extractive answer formats into a few-page setting. Long-document multimodal benchmarks then introduced documents whose total length routinely exceeds backbone context windows and whose evidence requires cross-

Dataset	Pages	Type	Venue	Domain	# Docs	Avg Pages	# Q/A	Multi/S Hop
DocVQA (2021)	SP	PT+BM	WACV	Industry	12767	–	50,000	S
FeTaQA (2022)	SP	PT+BM	TACL	Wikipedia tables	10330	–	10,330	M
InfographicVQA (2022)	SP	PT+BM	WACV	Infographics	5,485	–	30,035	M+S
PDF-VQA Task A (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	81,085	S
PDF-VQA Task B (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	53,872	S
TabFact (2020)	SP	PT+BM	ICLR	Wikipedia tables	16573	–	118,275	M+S
TextVQA (2019)	SP	PT+BM	CVPR	Scene-text images	28408	–	45,336	S
VisualMRC (2021)	SP	PT+BM	AAAI	Webpages	10197	–	30,562	M+S
WikiTableQuestions (2015)	SP	PT+BM	ACL	Wikipedia tables	2108	–	22,033	M
ChartQA (2022)	SP	BM	ACL	Charts	20882	–	32,719	M+S
DocGenome (2024)	MP	PT+BM	NeurIPS	Academic Paper	500000	13	3000	M+S
DuReaderVis (2022)	MP	PT+BM	ACL	Cross-domain	158000	–	15,000	S
Kleister-Charity (2021)	MP	PT+BM	ICDAR	Financial	2778	22.19	21,612	S
Kleister-NDA (2021)	MP	PT+BM	ICDAR	Legal	540	5.98	2,160	S
M-LongDoc (2025)	MP	PT+BM	EMNLP	Cross-domain	180	210.8	851	S
MMVQA (2024)	MP	PT+BM	IJCAI	Biomedical	3146	9.6	262,928	M+S
MP-DocVQA (2023)	MP	PT+BM	PR	Industry	5928	8.27	46,176	S
MultiHiertt (2022)	MP	PT+BM	ACL	Financial reports	2513	~2.5	10,440	M
PaperPDF (2026)	MP	PT+BM	preprint	Academic Paper	–	–	1,110,000	M+S
PDF-VQA Task C (2023)	MP	PT+BM	ECML-PKDD	Biomedical	1147	~10	5,653	M
QASPER (2021)	MP	PT+BM	NAACL	Academic Paper	1585	~8	5,049	M
SlideVQA (2023)	MP	PT+BM	AAAI	Presentations	2619	20	14,484	M+S
MP-DocStruct1M (2025)	MP	PT	ACL	Cross-domain	–	–	1,113,259	S
OpenDocVQA (2025)	MP	SFT+BM	CVPR	Cross-domain	–	–	–	M+S
CoR-Dataset (2026)	MP	SFT	preprint	Cross-domain	–	–	26,088	M+S
Doc-750K (2025b)	MP	SFT	CVPR	Academic Paper	–	–	758,000	M+S
MHDocVQA (2025)	MP	SFT	CVPR	Cross-domain	–	9.5	38,020	M
MP-DocReason51K (2025)	MP	SFT	ACL	Cross-domain	–	–	51,754	M+S
BRIDGE (2026)	MP	BM	preprint	Academic Paper	262	–	11,857	M
DocBench (2024)	MP	BM	preprint	Cross-domain	229	–	1,102	S
DUDE (2023)	MP	BM	ICCV	Cross-domain	5019	5.72	41,541	M+S
FetaTab (2024)	MP	BM	NeurIPS	Wikipedia Articles	871	~11	1,016	S
LongDocURL (2025)	MP	BM	ACL	Cross-domain	396	85.6	2,325	M+S
M3DocVQA (2025)	MP	BM	CVPR	Wikipedia Articles	3368	12.2	2,441	M+S
MM-NIAH (2024b)	MP	BM	NeurIPS	Synthetic	–	–	16,486	M+S
MMDocIR (2025)	MP	BM	ACL	Cross-domain	313	65.1	1,658	M+S
MMLongBench (2025c)	MP	BM	NeurIPS	Cross-domain	–	–	13,331	M+S
MMLongBench-Doc (2024)	MP	BM	NeurIPS	Cross-domain	135	47.5	1,082	M+S
PaperTab (2024)	MP	BM	NeurIPS	Academic Paper	307	~11	393	M+S
PaperText (2024)	MP	BM	NeurIPS	Academic Paper	1087	~11	2,804	M+S
ViDoSeek (2025a)	MP	BM	EMNLP	Cross-domain	–	–	1,200	M+S
VisDoMBench (2025)	MP	BM	NAACL	Cross-domain	1277	19.11	2,271	M+S
VisR-Bench (2025)	MP	BM	ICLR	Cross-domain	226	–	471	M+S
DocStruct4M (2024)	MP+SP	PT+SFT	EMNLP	Cross-domain	–	–	4,000,000	M+S
TAT-DQA (2022)	MP+SP	PT+BM	ACM	Financial reports	2758	1.33	16,558	M
InstructDoc (2024)	MP+SP	SFT	AAAI	Cross-domain	–	–	–	M+S
Leopard-Instruct (2025)	MP+SP	SFT	TMLR	Cross-domain	–	–	925,000	M+S

Table 4: Datasets referenced in the MP-VRDU survey. **Pages**: document scope (MP = Multi-Page, SP = Single-Page, MP+SP = Mixed). **Type**: dataset role – PT (Pretraining), SFT (Supervised Fine-Tuning), BM (Benchmark), and combinations. **Hop**: question-hop type (Single, Multi, or both). '–' indicates not reported.

page integration over mixed text, tables, charts, and figures, including MMLongBench-Doc (Ma et al., 2024), LongDocURL (Deng et al., 2025), DocBench (Zou et al., 2024), MMDocIR (Dong et al., 2025), and M-LongDoc (Chia et al., 2025), while retrieval-oriented benchmarks such as UDA-Benchmark (Hui et al., 2024), DocR1 (Xiong et al., 2026), and DocLens (Zhu et al., 2026) additionally score evidence-page selection and layout-level localisation alongside answer accuracy. The most recent additions push further into cross-document settings such as M3DocRAG (Cho et al., 2024), ViDoRAG (Wang et al., 2025a), VDocRAG (Tanaka et al., 2025), and VisDoM (Suri et al., 2025), while also expanding beyond the academic, financial, and slide-deck domains that dominated earlier work through datasets such as SVRAG (Chen et al., 2025). Despite this proliferation, evaluation in practice has consolidated around a small set of long-document and mid-length benchmarks that recur in most reported comparisons and serve as the de-facto standards for cross-model assessment.

E Additional Resources

E.1 Reported Performance by Architecture

Table 5 summarises reported scores on the five most-adopted multi-page benchmarks, grouped by architectural family. It is included for coverage rather than ranking. The matrix is sparse and metric conventions diverge across MP-DocVQA, DUDE, LongDocURL, MMLongBench-Doc, and SlideVQA, so scores are not comparable across columns; for the same benchmark, differences in backbone scale, proprietary-LLM dependency, evidence budget, and the training-versus-zero-shot

Model	Variant	MP-DocVQA	DUDE	LongDocURL	MMLongBench-Doc		SlideVQA	
		ANLS	ANLS	Accuracy	Accuracy	F1	EM	F1
General Domain MLLM								
Claude-3-Opus	Claude-3-Opus	–	–	–	17.4	18.1	–	–
Gemini-1.5-Pro	Gemini-1.5-Pro	–	–	50.9	28.2	20.6	–	–
GPT-4o	GPT-4o (LVLM mode)	–	–	64.5	42.8	44.9	–	–
GPT-4V	GPT-4V(ision)	–	–	–	32.4	31.2	–	–
InternVL-Chat-v1.5	InternVL-Chat-v1.5-26B	–	–	–	14.6	13.0	–	–
LLaVA-OneVision-7B	LLaVA-OneVision-Chat-7B	–	–	25.0	–	–	–	–
MiniCPM-Llama3-V2.5-8B	MiniCPM-Llama3-V2.5	–	–	–	8.5	8.6	–	–
Qwen2-VL-7B	Qwen2-VL-7B	–	–	30.6	–	–	–	–
Page-to-Document Encoding Models								
Arctic-TILT	Arctic-TILT	81.2	58.1	–	25.8	–	55.1	–
GRAM	GRAM-859M	80.32	51.15	–	–	–	–	–
Hi-VT5	Hi-VT5 (multipage)	62.01	–	–	–	–	–	–
RM-T5	RM-T5	64.01	–	–	–	–	–	–
MLLM-Centric Adaptation Models								
DocR1	DocR1	87.45	54.39	–	–	–	–	71.96
DocSeeker	DocSeeker (full SFT + EviGRPO)	86.2	57.4	51.7	40.1	–	–	77.1
Leopard	Leopard-LLaVA-Pro	70.6	43.82	–	–	–	–	37.51
mPLUG-DocOwl2	DocOwl2-8B	69.42	46.77	–	–	–	–	–
URaG	URaG-7B	88.2	57.6	52.2	33.8	32.8	72.1	–
Docopilot	Docopilot-8B	81.3	–	–	28.8	23	–	–
Texthawk2	Texthawk2	–	–	–	–	–	–	–
DocSLM	DocSLM-2B	70	47.6	–	–	–	–	–
InstructDoc	InstructDoc (finetuned)	–	46.8	–	–	–	37.7	47.3
LayTokenLLM	LayTokenLLM-8B	74.31	52	–	–	–	–	–
CoR	Qwen2.5-VL-CoR-7B	–	–	51.5	37.4	36	–	–
Retrieval-Augmented Pipelines								
AVIR	AVIR	84.58	–	–	–	–	60.3	68.9
DFVC	DFVC (ColQwen backbone)	–	–	–	–	–	–	–
DREAM	DREAM† (InternVL2-4B + retrieved text)	85.72	52.14	–	27.3	28.6	–	–
M3DocRAG	M3DOCRAG (ColPali + Qwen2-VL 7B)	84.44	–	–	21	22.6	–	–
MM-R5	MM-R5 (Qwen2.5-VL-7B SFT+GRPO)	–	–	–	–	–	–	–
MoLoRAG	MoLoRAG+ (Qwen2.5-VL-7B)	–	–	51.85	41.01	–	–	–
Self-Attention Scoring	Pix2Struct-SA proposed	61.99	–	–	–	–	–	–
SLEUTH	SLEUTH (Qwen3-VL-8B + ColPali Top-5)	–	–	59.96	52.77	–	–	–
SV-RAG	SV-RAG-InternVL2† R5 (4B)	71.0	45.0	–	23.0	24.2	31.98	–
VDocRAG	VDocRAG	–	44	–	–	–	–	42
CREAM	CREAM	65.32	52.46	–	–	–	–	–
MHier-RAG	MHier-RAG (Qwen-turbo)	–	–	55.7	52.3	46	–	–
MLDocRAG	MLDocRAG	–	–	50.8	47.9	–	–	–
MultiDocFusion	MultiDocFusion	–	–	–	–	–	–	–
M2RAG	M[2]RAG top-5	–	–	57.68	39.9	–	–	–
MDocAgent	MDocAgent (top-4)	–	–	57.8	31.5	–	–	–
PDF-WuKong	PDF-WuKong	76.9	56.1	–	–	–	–	–
RAG-DocVQA	RAG-Qwen2.5-VL	73.7	39.9	–	–	–	–	–
VisDoMRAG	VisDoMRAG (ChatGPT-4o + ColPali + BGE)	–	–	–	–	–	–	67.22
KGP	KGP	–	–	–	–	–	–	–
Adaptive-Trajectory Pipelines								
Doc-V*	Doc-V* (GRPO)	86.2	64.5	56.3	42.1	–	–	77.2
DocReact	DocReact (w/ ColPali)	–	–	–	38.29	38.07	54.87	55.04
MACT	MACT-MiMo-VL-Series-28B	–	70.8	–	–	47.4	–	83.9
VRAG-RL	VRAG-RL (Qwen2.5-VL-7B-Instruct)	–	–	–	–	–	–	–
DocLens	DocLens (Gemini-2.5-Pro)	–	–	–	67.6	–	–	–
DMAP	DMAP Ours-Top4 (GPT-4o backbone)	–	–	60.7	43.2	–	–	–
DocAgent	DocAgent (Claude-3.5-Sonnet)	–	–	–	57.3	54.1	–	–
DocDancer	DocDancer (GPT-5.2)	–	–	–	57	56.8	–	–
MM-Doc-R1	MM-Doc-R1 (Qwen3-8B + SPO)	–	–	–	49.7	46.1	–	–
SimpleDoc	SimpleDoc (3.5 pages)	–	–	72.3	60.58	–	–	–
ViDoRAG	ViDoRAG	–	–	–	–	–	–	–

Table 5: Performance of MP-VRDU models grouped by architecture category. Reported metrics follow the source papers and are not always directly comparable. '–' indicates not reported.

distinction (Table 8) mean that a higher score does not by itself indicate a better architecture. Several of the strongest systems orchestrate closed-source frontier backbones of undisclosed scale, so their scores cannot be attributed to architecture alone. Read with these caveats, the table supports one robust cross-family observation: coverage tracks cost structure, with page-to-document and MLLM-centric systems confined to mid-length benchmarks and selective-evidence families dominating the long-document frontier.

E.2 General Domain MLLM for MP-VRDU

Frontier general-domain MLLMs handle short multi-page inputs well when the document fits in context, but the multi-page setting introduces three pressures absent in single-page VRDU: context-window overrun, attention dilution over sparsely distributed evidence (Liu et al., 2024), and a per-page reading fidelity versus whole-document coverage trade-off under a fixed token budget. Table 5 shows that even the strongest frontier MLLMs trail specialised MP-VRDU systems by a wide margin

Dataset	PT	SFT	BM	SFT ∩ BM
MP-DocVQA	0	19	20	16
DUDE	0	16	20	12
MMLongBench-Doc	0	3	25	2
DocVQA	0	10	14	8
SlideVQA	0	7	17	6
InfographicVQA	0	9	12	7
LongDocURL	0	3	14	2
ChartQA	0	5	7	4
VisualMRC	0	4	5	2
FetaTab	0	0	6	0
PaperTab	0	0	6	0
TabFact	0	2	3	1

Table 6: Top 12 datasets by total appearance count across the surveyed corpus, with usage counts broken down by role. **SFT ∩ BM** counts the papers that use the dataset in *both* fine-tuning and benchmarking.

on MMLongBench-Doc and LongDocURL, motivating the specialised architectures examined in §3.

E.3 Contamination Matrix

Table 6 quantifies the contamination risk discussed in §8. Overlap between fine-tuning and benchmarking is highest on mid-length corpora: 16 of the 19 papers that fine-tune on MP-DocVQA also evaluate on it, with comparable patterns for DUDE, SlideVQA, and InfographicVQA. Long-document benchmarks show smaller overlap so far, though their fine-tuning adoption is rising.

E.4 Open-Source Code Releases

Table 7 lists the surveyed MP-VRDU models that release implementations through a public GitHub repository: 16 papers link such a repository directly. Three further systems (DocLens, RAG-DocVQA, VDocRAG) distribute through `github.io` project pages, MACT announces a forthcoming release without a URL, and the remaining papers do not announce open-source releases.

E.5 Training Status by Component

Table 8 records which components are trained, frozen, or not applicable in each surveyed system across pretraining, instruction-tuning, and fine-tuning stages. Three patterns recur. *End-to-end* systems train multiple components jointly and dominate page-to-document encoding models. *Partially trained* systems freeze the heavy backbone and update only a light adaptor or projection layer, the prevalent pattern across retriever-generator and retrieval-augmented backbone variants. *Training-free* systems perform no parameter updates and dominate adaptive-trajectory pipelines built on prompted, multi-agent orchestration of frozen MLLMs. This view also clarifies why some scores in Table 5 are not directly comparable across the training/zero-shot boundary.

E.6 Model Components Details

Table 9 lists the concrete components of each surveyed system, organised by architectural family. The four families assemble these parts in characteristically different ways. Page-to-document transformers pair smaller LLM backbones (T5 variants, DocFormerv2) with custom layout-aware projectors. MLLM-centric adaptation models reuse standard vision-language stacks such as Qwen2.5-VL, mPLUG-Owl2, and SigLIP variants, and update only light projection or compression modules. Retriever-augmented pipelines decouple a frozen retriever (ColPali, ColQwen, VisRAG) from a frozen LVLM and train only the scorer, fuser, or reranker. Adaptive-trajectory pipelines mostly orchestrate frozen frontier VLMs, with parameter updates restricted to a few systems that fine-tune the controller via SFT or GRPO.

Model	GitHub Repository
Arctic-TILT 2025	https://github.com/Snowflake-Labs/arctic-tilt
AVIR 2025	https://github.com/Li-yachuan/AVIR-main
DMAP 2026	https://github.com/Forlorin/DMAP
DocAgent 2025	https://github.com/lisun-ai/DocAgent
Docopilot 2025b	https://github.com/OpenGVLab/Docopilot
DocSeeker 2026	https://github.com/yh-hust/DocSeeker
DocSLM 2026	https://github.com/Tanveer81/DocSLM
Hi-VT5 2023	https://github.com/rubenpt91/MP-DocVQA-Framework
InstructDoc 2024	https://github.com/nttmdlab-nlp/InstructDoc
KGP 2024c	https://github.com/YuWVandy/KG-LLM-MDQA
Leopard 2025	https://github.com/tencent-ailab/Leopard
MDocAgent 2025	https://github.com/aiming-lab/MDocAgent
MM-R5 2025	https://github.com/i2vec/MM-R5
MoLoRAG 2025b	https://github.com/WxxShirley/MoLoRAG
mPLUG-DocOwl2 2025	https://github.com/X-PLUG/mPLUG-DocOwl
PDF-WuKong 2026	https://github.com/yhhust/PDF-Wukong
Self-Attention Scoring 2024	https://github.com/leitro/SelfAttnScoring-MPDocVQA
SimpleDoc 2025	https://github.com/ag2ai/SimpleDoc
URaG 2026	https://github.com/shi-yx/URaG
ViDoRAG 2025a	https://github.com/Alibaba-NLP/ViDoRAG
VisDoMRAG 2025	https://github.com/MananSuri27/VisDoM
VRAG-RL 2025b	https://github.com/Alibaba-NLP/VRAG

Table 7: Open-source GitHub repositories released for surveyed MP-VRDU models.

Model	Arch.	Vision Encoder			LLM Backbone			Adaptors		
		PT	IT	FT	PT	IT	FT	PT	IT	FT
End-to-End										
Arctic-TILT	PDE	✓	–	✓	✓	–	✓	✓	–	✓
GRAM	PDE	–	–	✓	–	–	✓	–	–	✓
Docopilot	MLLM	–	✓	–	–	✓	–	–	✓	–
DocR1	MLLM	–	–	✓	–	–	✓	–	–	–
DocSeeker	MLLM	–	✓	✓	–	✓	✓	–	✓	✓
DocSLM	MLLM	✓	×	×	×	✓	✓	✓	✓	✓
mPLUG-DocOwl2	MLLM	✓	–	×	×	–	✓	✓	–	✓
Texthawk2	MLLM	✓	✓	–	✓	✓	–	✓	✓	–
PDF-WuKong	RA	–	–	✓	–	–	✓	–	–	✓
Doc-V*	AT	–	×	×	–	✓	✓	–	×	×
DocDancer	AT	–	×	–	–	✓	–	–	–	–
MACT	AT	–	✓	✓	–	✓	✓	–	–	–
VRAG-RL	AT	–	✓	✓	–	✓	✓	–	✓	✓
Partially Trained										
Hi-VT5	PDE	×	–	×	✓	–	✓	✓	–	✓
RM-T5	PDE	–	–	×	✓	–	✓	✓	–	✓
CoR	MLLM	–	×	×	–	×	✓	–	✓	✓
InstructDoc	MLLM	–	×	×	–	×	×	–	✓	✓
LayTokenLLM	MLLM	–	–	–	×	×	–	✓	✓	–
Leopard	MLLM	×	×	–	×	✓	–	✓	✓	–
URaG	MLLM	–	–	×	–	–	×	–	–	✓
AVIR	RA	–	–	×	–	–	×	–	–	✓
CREAM	RA	–	–	×	–	–	×	–	–	✓
DFVC	RA	–	–	×	–	–	×	–	–	✓
DREAM	RA	–	–	×	–	–	×	–	–	✓
KGP	RA	–	–	–	✓	✓	–	–	✓	–
M2RAG	RA	–	–	×	–	–	×	–	–	✓
MLDocRAG	RA	–	–	✓	–	–	✓	–	–	–
MM-R5	RA	–	×	×	–	✓	✓	–	✓	✓
MoLoRAG	RA	–	–	✓	–	–	✓	–	–	–
MultiDocFusion	RA	–	–	–	–	✓	–	–	✓	–
RAG-DocVQA	RA	–	–	✓	–	–	✓	–	–	✓
Self-Attention Scoring	RA	–	–	×	–	–	×	–	–	✓
SV-RAG	RA	–	–	×	–	–	×	–	–	✓
VDocRAG	RA	×	–	×	×	–	×	✓	–	✓
MM-Doc-R1	AT	–	–	×	–	–	✓	–	–	✓
Training-Free										
M3DocRAG	RA	–	–	–	–	–	–	–	–	–
MDocAgent	RA	–	–	–	–	–	–	–	–	–
MHier-RAG	RA	–	–	–	–	–	–	–	–	–
SLEUTH	RA	–	–	–	–	–	–	–	–	–
VisDoMRAG	RA	–	–	–	–	–	–	–	–	–
DMAP	AT	–	–	–	–	–	–	–	–	–
DocAgent	AT	–	–	–	–	–	–	–	–	–
DocLens	AT	–	–	–	–	–	–	–	–	–
DocReact	AT	–	–	–	–	–	–	–	–	–
SimpleDoc	AT	–	–	–	–	–	–	–	–	–
ViDoRAG	AT	–	–	–	–	–	–	–	–	–

Table 8: Training status of MP-VRDU model components grouped by training category. ✓denotes trained parameters, ×denotes frozen parameters, and '–' indicates not reported / not applicable. Architectural family abbreviations: PDE = Page-to-Document Encoding Models; MLLM = MLLM-Centric Adaptation Models; RA = Retrieval-Augmented Pipelines; AT = Adaptive-Trajectory Pipelines.

Table 9: Per-model components of the surveyed MP-VRDU systems. Resolution lists the input image resolution or page-rendering setting; *dynamic* denotes encoders that accept variable-resolution input. '–' indicates not reported not applicable.

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
Page-to-Document Encoding Models				
Arctic-TILT	U-Net	400k tokens / 500p	T5-Large	TILT fusion
GRAM	DocFormerv2	800 tok/page train, 8000 test	DocFormerv2	–
Hi-VT5	DiT	224x224 (DiT default), 20480 max tokens	T5	–
RM-T5	DiT	512 tok/page max	T5-base	–
MLLM-Centric Adaptation Models				
DocR1	Qwen2.5-VL	dynamic, max 1024 patches	Qwen2.5-VL-7B	Frozen Qwen MLP projector
DocSeeker	Qwen2.5-VL ViT	1024x784 eval, EGRA training	Qwen2.5-VL-7B-Instruct	–
Leopard	SigLIP-SO400M	364x364 / 980x980, max 50 sub-imgs	LLaMa-3.1-8B / Mistral-7B	MLP / Idefics2 resampler
mPLUG-DocOwl2	ViT-L/14	504x504, max 12 crops	mPLUG-Owl2	H-Reducer + DocCompressor
URaG	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-3B / Qwen2.5-VL-7B	Cross-modal retrieval module (2 linear + GELU) + LoRA
Docopilot	InternViT-300M	448 tile, max 24 tiles	InternLM2-1.8B / InternLM2.5-7B	2-layer MLP
Texthawk2	SigLIP-SO400M	224x224 base, max 72 sub-imgs	Qwen2-7B-Instruct	QPN + ReSA
DocSLM	SigLIP2	dynamic 384-1152, 576 tok/page	Qwen2.5-1.5B	OCR/Vision compressor + MLP
InstructDoc	CLIP	224x224	FlanT5 / BLIP2	Doc-former + proj
LayTokenLLM	–	–	LLaMA3-8B / Qwen1.5-7B	Layout tokenizer + LoRA
CoR	Qwen2.5-VL	dynamic, 32k max	Qwen2.5-VL-7B	Frozen Qwen MLP projector + LoRA
Retrieval-Augmented Pipelines				
AVIR	Pix2Struct	dynamic, 768-5120 patches	Qwen2.5-VL-3B-AWQ	Frozen Qwen MLP projector
DFVC	VisRAG, ColQwen	dynamic	Qwen2-VL-2B-Instruct	Cross-page MLP
DREAM	InternVL2 ViT	448 tile, max 24 tiles	InternVL2-4B	Cross-page MoE attention connector + LoRA
M3DocRAG	ColPali / ColQwen	144 DPI, ~1700px wide	Qwen2-VL-7B / InternVL2-8B / Idefics2-8B / Idefics3-8B	–
MM-R5	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-7B	–
MoLoRAG	ColPali	448x448	Qwen2.5-VL 3B/7B / Deepseek-VL-16B / LlaVa-Next-7B	Frozen base projector
Self-Attention Scoring	Pix2Struct	2048 patches	Pix2Struct	–
SLEUTH	ColPali, Qwen3-VL ViT	448x448	Qwen3-VL-8B / GLM-4.1V-Thinking-8B / Gemini-2.5-Flash	–
SV-RAG	InternVL2 / PaliGemma / Phi-3-vision	dynamic (InternVL2 default), 448x448	InternVL2-4B / PaliGemma-3B / Phi-3-vision-4B	Dual LoRA adapters (retrieval + QA) + Col-projection
VDocRAG	Phi3V-4.2B	336x336 dynamic, up to 1344x1344	Phi3V-4.2B	LoRA + MLP
CREAM	Pix2Struct	2048 patches	LLaMA2-7B	Attn pooling + MLP + LoRA
KGP	–	–	T5-large, LLaMa-7B, MDR (RoBERTa), ChatGPT	LoRA
MHier-RAG	Qwen-VL-Plus	dynamic	Qwen-VL-Plus / GPT-3-turbo / Qwen-turbo / GPT-4o / DeepSeek-chat / ERNIE-turbo	–
MLDocRAG	LVLM Qwen2.5-VL	dynamic	Qwen2.5-VL-32B-Instruct, Qwen2.5-VL-7B, Qwen2.5-72B	Frozen Qwen MLP projector
MultiDocFusion	–	–	Mistral-8B / Llama-3.2-3B / Qwen-2.5-3B / Qwen-2.5-7B	LoRA

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Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
M2RAG	VisRAG-Ret	dynamic	Qwen2.5-7B-Instruct, Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B	Frozen Qwen MLP projector + LoRA + modal fuser
MDocAgent	ColPali / ColQwen2	144 DPI	LLaMA-3.1-8B-Instruct, Qwen2-VL-7B-Instruct	–
PDF-WuKong	IXC2-VL-4KHD	dynamic 336px-4K, max 16 tiles	IXC2-VL-4KHD	Sparse sampler
RAG-DocVQA	DiT / Pix2Struct	2048 patches	VT5 / Qwen2.5-VL-7B-Instruct	LoRA
VisDoMRAG	ColPali / ColQwen2	448x448 (ColPali), dynamic	Qwen2-VL-7B-Instruct / Gemini-1.5-Flash / GPT-4o	–
Adaptive-Trajectory Pipelines				
Doc-V*	Qwen2.5-VL ViT	1024x768 / 256x256 thumb	Qwen2.5-VL-7B-Instruct	Frozen MLP
DocReact	ColPali, VisRAG-Ret	448x448 (ColPali default) / dynamic	GPT-4o	–
MACT	Qwen2.5-VL-7B-Instruct / MiMo-VL-7B-SFT / InternVL3-9B	dynamic	Qwen2.5-VL-7B / MiMo-VL-7B-SFT / InternVL3-9B, Qwen2.5-7B/3B / MiMo-7B-SFT / InternVL3-8B/2B	–
VRAG-RL	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-3B-Instruct / Qwen2.5-VL-7B-Instruct	–
DocLens	frontier VLMs	dynamic	Gemini-2.5-Pro / Gemini-2.5-Flash / Claude-4-Sonnet	–
DMAP	ColPali	dynamic	GPT-4o / Qwen-plus	–
DocAgent	–	144 DPI	Gemini 2.0 Flash / GPT-4o / Claude Sonnet 3.5	–
DocDancer	Qwen3-VL-235B-A22B-Instruct	144 DPI	Qwen3-4B/30B-A3B	–
MM-Doc-R1	Qwen2.5-VL ViT	dynamic	Qwen3-4B / Qwen3-8B	–
SimpleDoc	ColQwen-2.5	dynamic	Qwen2.5-VL-32B-Instruct, Qwen3-30B-A3B	–
ViDoRAG	Nv-embed-V2, ColQwen2	dynamic	Qwen2.5-VL / LLaMA3.2-Vision / GPT-4o	–

Table 10: Per-dataset usage across the surveyed corpus, '-' indicates no recorded usage. Only representative datasets from Table 4 used by at least 1 paper are included, pretraining datasets are deliberately excluded.

Dataset	Fine-Tuning	Benchmarking
ChartQA	Docopilot, DocR1, mPLUG-DocOwl2, PDF-WuKong, VDocRAG	CoR, Docopilot, DocR1, MACT, PDF-WuKong, Texthawk2, VDocRAG
CoR-Dataset	CoR	-
Doc-750K	Docopilot	-
DocBench	-	DocAgent, DocDancer, M2RAG
DocGenome	mPLUG-DocOwl2	-
DocStruct4M	mPLUG-DocOwl2	-
DocVQA	Arctic-TILT, CREAM, Docopilot, DocR1, LayTokenLLM, mPLUG-DocOwl2, PDF-WuKong, Self-Attention Scoring, SV-RAG, VDocRAG	Arctic-TILT, CoR, CREAM, DocR1, DocReact, Hi-VT5, LayTokenLLM, Leopard, MACT, mPLUG-DocOwl2, PDF-WuKong, Self-Attention Scoring, SV-RAG, Texthawk2
DUDE	Arctic-TILT, CREAM, Doc-V*, DocDancer, Docopilot, DocR1, DocSeeker, DREAM, GRAM, LayTokenLLM, MM-R5, mPLUG-DocOwl2, PDF-WuKong, RAG-DocVQA, URaG, VDocRAG	Arctic-TILT, AVIR, CREAM, Doc-V*, DocR1, DocSeeker, DocSLM, DREAM, GRAM, LayTokenLLM, Leopard, M3DocRAG, MACT, MLDocRAG, MultiDocFusion, PDF-WuKong, RAG-DocVQA, SV-RAG, URaG, VDocRAG
FetaTab	-	DMAP, MDocAgent, MoLoRAG, SimpleDoc, SLEUTH, VisDoMRAG
InfographicVQA	Arctic-TILT, CREAM, Docopilot, DocR1, mPLUG-DocOwl2, PDF-WuKong, RAG-DocVQA, SV-RAG, VDocRAG	Arctic-TILT, CREAM, Docopilot, DocR1, Hi-VT5, InstructDoc, M3DocRAG, MACT, PDF-WuKong, RAG-DocVQA, Texthawk2, VDocRAG
InstructDoc	InstructDoc	InstructDoc

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Dataset	Fine-Tuning	Benchmarking
Kleister-Charity	Arctic-TILT, DocR1, mPLUG-DocOwl2	Arctic-TILT
Kleister-NDA	Arctic-TILT	Arctic-TILT
Leopard-Instruct	Leopard	-
LongDocURL	DocDancer, MM-Doc-R1, MoLoRAG	CoR, DMAP, Doc-V*, DocDancer, DocLens, DocSeeker, M2RAG, MDocAgent, MHier-RAG, MLDocRAG, MoLoRAG, SimpleDoc, SLEUTH, URaG
M3DocVQA	-	M3DocRAG
MHDocVQA	VDocRAG	-
MM-NIAH	-	Docopilot
MMDocIR	-	MM-R5
MMLongBench-Doc	DFVC, MoLoRAG, VRAG-RL	Arctic-TILT, CoR, DMAP, Doc-V*, DocAgent, DocDancer, DocLens, Docopilot, DocReact, DocSeeker, DocSLM, DREAM, M2RAG, M3DocRAG, MACT, MDocAgent, MHier-RAG, MLDocRAG, MM-Doc-R1, MoLoRAG, SimpleDoc, SLEUTH, SV-RAG, URaG, VRAG-RL
MP-DocReason51K	mPLUG-DocOwl2	-
MP-DocVQA	CREAM, DFVC, Doc-V*, Docopilot, DocR1, DocSeeker, DocSLM, DREAM, GRAM, Hi-VT5, LayTokenLLM, M2RAG, MM-R5, mPLUG-DocOwl2, PDF-WuKong, RAG-DocVQA, RM-T5, Self-Attention Scoring, URaG	Arctic-TILT, AVIR, CREAM, DFVC, Doc-V*, DocR1, DocSeeker, DocSLM, DREAM, GRAM, Hi-VT5, LayTokenLLM, Leopard, M3DocRAG, mPLUG-DocOwl2, PDF-WuKong, RAG-DocVQA, RM-T5, Self-Attention Scoring, URaG
MultiHiertt	DocR1, Leopard	DocR1, Leopard
OpenDocVQA	-	VDocRAG
PaperPDF	PDF-WuKong	PDF-WuKong
PaperTab	-	DMAP, MDocAgent, MoLoRAG, SimpleDoc, SLEUTH, VisDoMRAG
PaperText	-	DMAP, MDocAgent, SLEUTH
SlideVQA	DocR1, MM-R5, SV-RAG, URaG, VDocRAG, ViDoRAG, VRAG-RL	Arctic-TILT, AVIR, Doc-V*, DocR1, DocReact, DocSeeker, Leopard, M2RAG, MACT, MDocAgent, MHier-RAG, MLDocRAG, SV-RAG, URaG, VDocRAG, ViDoRAG, VRAG-RL
TabFact	DocR1, mPLUG-DocOwl2	DocR1, InstructDoc, Texthawk2
TAT-DQA	Arctic-TILT, MM-R5, SV-RAG	Arctic-TILT
TextVQA	DocR1, mPLUG-DocOwl2	DocR1, Leopard, Texthawk2
ViDoSeek	VRAG-RL	ViDoRAG, VRAG-RL
VisDoMBench	-	VisDoMRAG
VisR-Bench	-	SV-RAG
VisualMRC	CREAM, DocR1, mPLUG-DocOwl2, VDocRAG	CREAM, DocR1, Hi-VT5, InstructDoc, MACT
WikiTableQuestions	DocR1, mPLUG-DocOwl2	DocR1, Texthawk2